
APM GROUP PRIVATE LIMITED

APM NEXUS

INTEGRATED VEHICLE INTELLIGENCE PLATFORM

SPECIFICATION SHEET

6-in-1 Integrated System

AIS 140 Government Compliant • AI Edge Computing • Real-Time Telematics

- GNSS Tracking & Compliance (Optional)
- RK3568 AI Edge Processing
- Ultrasonic Reverse Parking
- Payload / Load Monitoring
- Automatic Headlamp Dimming
- Front & IR Night Vision Rear Camera

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SYSTEM OVERVIEW

The APM Nexus is a 6-in-1 integrated vehicle intelligence platform built around the Rockchip RK3568 AIoT processing board. It unifies GPS/GNSS tracking, AI edge computing, ultrasonic reverse parking detection, payload monitoring, automatic headlamp dimming, and a rear-view camera into a single cohesive system — managed through one processing hub, one wiring architecture, and one fleet management interface.

Engineered for heavy commercial vehicles, buses, and large-scale fleet operations, the APM Nexus eliminates the complexity of multiple standalone aftermarket devices by consolidating all functions into one platform with shared power, shared connectivity, and centralized AI-driven data fusion.

KEY DIFFERENTIATORS

- ✓ Single RK3568 SoC hub processes all sensor data, camera feeds, and telemetry
- ✓ One 4G/5G modem transmits consolidated data stream
- ✓ One power harness and wiring architecture for entire system
- ✓ On-device AI inference (1 TOPS NPU) for real-time analytics at the edge
- ✓ AIS 140 regulatory compliance built-in with Tri-IP transmission
- ✓ Single OTA firmware update covers entire platform
- ✓ Dramatic reduction in installation time, failure points, and total cost of ownership

Target Applications

- Mining vehicles
- Public transport buses (state and private operators)
- School and college transport fleets
- Commercial logistics — trucks, tankers, trailers
- Taxi and cab aggregators
- Smart city ITS deployments

CENTRAL PROCESSING UNIT — RK3568 AIOT BOARD

The RK3568 system-on-chip serves as the central brain of APM Nexus — a quad-core ARM Cortex-A55 processor with integrated GPU, NPU, VPU, and ISP enabling edge computing, AI inference, video analytics, and sensor fusion directly within the vehicle.

Parameter	Specification
Processor (SoC)	Rockchip RK3568, 22nm, Quad-core Cortex-A55 @ 2.0 GHz
Architecture	64-bit ARMv8-A
GPU	Mali-G52 2EE; OpenGL ES 1.1/2.0/3.2, OpenCL 2.0, Vulkan 1.1
NPU	RKNN NPU, 1.0 TOPS; INT8/INT16/FP16 inference
AI Frameworks	TensorFlow Lite, ONNX, OpenCV 3.43+, RKNN Toolkit
Memory	2GB / 4GB LPDDR4
Storage	16GB / 32GB eMMC 5.1; TF card up to 256GB
Video Decode	H.264/H.265/VP8/VP9 up to 4K@60fps
Video Encode	H.264/H.265 up to 1080P@60fps
ISP	8M ISP, HDR, dual camera input (MIPI CSI-2)
Display 1	HDMI 2.0 (up to 4K@60fps)
Display 2	MIPI DSI (up to 1920x1080@60fps)
Display 3	LVDS dual-channel (up to 1920x1080@60fps)
Ethernet	2x Gigabit (10/100/1000 Mbps)
USB	1x USB 3.0 Host, 2x USB 2.0 Host, 1x USB-C OTG
PCIe	1x PCIe 3.0 x1, 1x PCIe 2.1 x1
SATA	SATA 3.0 (6 Gbps) for NVR/storage
CAN Bus	2x CAN-FD (ISO 11898, up to 5 Mbps)
Serial Ports	Multiple UART (RS232/RS485/TTL configurable)
GPIO / SPI / I2C	Configurable GPIO; SPI x2; I2C x5
Audio	I2S, onboard MIC, headphone jack, speaker output
Wireless	M.2/mPCIe for 4G/5G; optional Wi-Fi 5 + BT 5.0
RTC	Onboard with battery backup
Power Input	12V DC (9–36V with onboard regulation)
Power Consumption	5W idle / 10W full load
Operating Temp	-20°C to +70°C (industrial grade)
OS Support	Linux (Buildroot/Debian/Ubuntu), Android 11/12, RTOS
SoM Dimensions	50 x 42 mm
Certifications	CE, FCC, RoHS

GNSS TRACKING & AIS 140 REGULATORY COMPLIANCE

The integrated AIS140-compliant GNSS module provides real-time vehicle tracking with multi-constellation satellite support, 4G LTE connectivity, and dual-IP transmission for simultaneous regulatory and fleet management reporting. ARAI, ICAT, and DIMTS certified.

Parameter	Specification
GNSS Chipset	QUECTEL
Constellations	GPS / GLONASS / NavIC (IRNSS) / GAGAN (SBAS)
Position Accuracy	≤ 2.5m CEP (open sky)
Cold Start	≤ 35 seconds
Hot Start	≤ 1 second
Cellular	4G LTE Cat-1 with 2G fallback / GSM Class 12
Protocols	TCP/IP, SMS, MQTT
SIM	eSIM
Transmission	Tri-IP (regulatory server+ EMR Server + fleet platform)
Data Logging	40,000+ records offline with auto-sync
Operating Voltage	8V – 32V DC (12V/24V compatible)
Battery Backup	Li-Polymer 3.7V, 1200mAh; min. 4 hrs @ 60s interval
Accelerometer	3-axis MEMS (±2g/±4g/±8g configurable)
Gyroscope	3-axis (tilt / rollover detection)
Digital Inputs	4 ch (Ignition, SOS, Door, configurable)
Analog Inputs	2 ch (0–5V for fuel/load sensor)
Serial Interface	1x RS232 (RFID, camera, display)
CAN Interface	J1939/OBD-II (optional variant)
SOS Button	Built-in panic button with emergency alert
Tamper Detection	Disconnect alert on housing breach
OTA Updates	Over-the-air firmware capability
Antenna	Internal GNSS + GSM; external SMA port
Ingress Protection	IP65
Dimensions	95 x 65 x 28 mm
Weight	≤ 150g (with battery)
Certifications	ARAI AIS 140

ULTRASONIC REVERSE PARKING SYSTEM

Five-sensor ultrasonic array with progressive 5-zone alerting, integrated with the rear-view camera for a unified reversing safety system. All data is fused on the RK3568 board for combined distance + video overlay on the in-cabin display.

Parameter	Specification
Sensor Technology	Ultrasonic, 58 kHz
Number of Sensors	5 units (rear bumper flush-mount)
Detection Range	0 – 3m standard; 0 – 5m extended (on order)
Detection Angle	X = Y = 60° – 80°
Response Time	0.05 seconds per sensor
Operating Voltage	12V / 24V DC
Operating Temp	-30°C to +70°C
Ingress Protection	IP67 (waterproof, dust-tight)
Connector Type	Aviation plug (vibration-resistant, locking)
EMI Protection	Shielded metal enclosure
Cable Length	sensor to ECU
Min. Mount Height	≥ 50 cm from ground
Sensor Clearance	≥ 30 cm around each sensor

Alarm Zone Scheme

Zone	Distance	Mode	Alert
01	0 – 30 cm	STOP	Continuous tone
02	30 – 100 cm	Alarming	Fast beep
03	100 – 200 cm	Caution	Slow beep
04	200 – 300 cm	Alerting	Visual only
05	> 300 cm	Safe	None

FRONT / REAR VIEW CAMERA SYSTEM

Heavy-duty IP68/IP69K camera with IR night vision, directly integrated with the parking sensor ECU and the RK3568 AI board for combined video overlay, pedestrian detection, and event recording.

Parameter	Specification
Image Sensor	1/3" CMOS (Color)
Resolution	720P / optional 1080P
Lens Angle	120° / 150° / 170° (selectable)
Video System	PAL / NTSC (auto-detect)
Video Output	Composite CVBS / AHD

IR Night Vision	18x IR LEDs, 10–15m range, auto-activate
Min. Illumination	0.1 Lux (0 Lux with IR on)
Operating Voltage	12V – 24V DC
Power Consumption	≤ 3W
Waterproof Rating	IP68 / IP69K
Vibration	10G (MIL-STD-810G)
Operating Temp	-30°C to +70°C
Housing	Die-cast aluminum + stainless steel bracket
Cable Length	Camera to ECU
Features	Mirror/Normal mode; parking guide overlay; auto-activate on reverse

PAYLOAD / LOAD MONITORING SENSOR

Rotary position sensor for leaf-spring suspension vehicles. Translates axle deflection into a 0–5V analog signal fed directly to the Nexus tracker, with the RK3568 board applying calibration tables for real-time weight estimation via linear interpolation.

Parameter	Specification
Sensor Type	Rotary position / Angle Variation Detection
Signal Output	0 – 5V DC analog
Accuracy	±2% F.S.
Detection Range	0° – 120°
Power Input	9V – 32V DC
No-Load Voltage	≈ 1.85V typical (vehicle-dependent)
Full-Load Voltage	≈ 3.44V typical (vehicle-dependent)
Operating Temp	-20°C to +85°C
Ingress Protection	IP67
Material	Aluminum alloy body + ABS housing
Mounting	Weld-mount backplane + angle iron to axle
Installation	Single sensor per axle/bogie sufficient
Integration	Connects to Nexus tracker analog input (Pin 7/8)

Wire Connection

Wire	Color	Function
Power	Red	Positive 12V / 24V DC
Ground	Black	Negative / GND
Signal	Blue	0 – 5V analog output (proportional to load)

AUTOMATIC HEADLAMP DIMMING — AD-100

Intelligent beam control using photosensitive array to detect oncoming headlights at 200–500m and auto-switch from high to low beam in ≤ 0.3 seconds. Digital output logs all events to the Nexus fleet platform for nighttime compliance reporting.

Parameter	Specification
Model	APM Nexus AD-100
Sensor Type	Photosensitive array (Photodiode + IR filter)
Detection Range	200 – 500 meters
Detection Angle	Horizontal: $\pm 15^\circ$; Vertical: $\pm 5^\circ$
Response Time	≤ 0.3 seconds
Operating Voltage	9V – 32V DC
Current Draw	≤ 50 mA standby; ≤ 200 mA relay active
Relay Output	1x SPDT 30A
Digital Output	Open-collector
Sensitivity	3-level potentiometer (Low / Medium / High)
Manual Override	Dedicated toggle switch included

Wire Connection

Wire Color	Function
Red	Power +VE (12V/24V)
Black	Ground (GND)
Yellow	High beam input (from stalk switch)
White	Low beam relay output
Green	High beam relay output
Blue	Digital output to Nexus tracker
Orange	Manual override switch input

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